

Intelligent BCP/DR Orchestration & Compliance Control Plane

Deterministic Recovery Authority • Dependency-Aware State • Checkpointed Execution • Failure-Domain Isolation

Principal AI Architect — AI Platforms & Distributed Systems

Capital Markets / Trade Operations • AWS EKS / RDS / S3 • 21-person cross-functional engineering team • 11-month program

ARCHITECTURE-FIRST • PRODUCTION-ORIENTED • ANONYMIZED

Production evidence	Documented result
RTO	Sub-5-minute recovery
RPO	Near-zero
Data integrity	Zero data loss
Automated validation	95%+
Recovery success	99%+

01 • EXECUTIVE ARCHITECTURE MANDATE

The architecture re-engineered BCP/DR recovery around a deterministic control plane. The problem was not simply how to automate recovery commands; it was how to represent dependency state, recovery progress, replay safety, validation and evidence under distributed partial failure.

Control-plane authority

The control plane owns incident declaration, policy validation, dependency-aware planning, checkpoint progression, recovery state transitions and validation. Infrastructure provisioning and downstream service execution remain separate boundaries.

The documented production contribution is the deterministic recovery model; no unsupported autonomous AI decision authority is asserted.

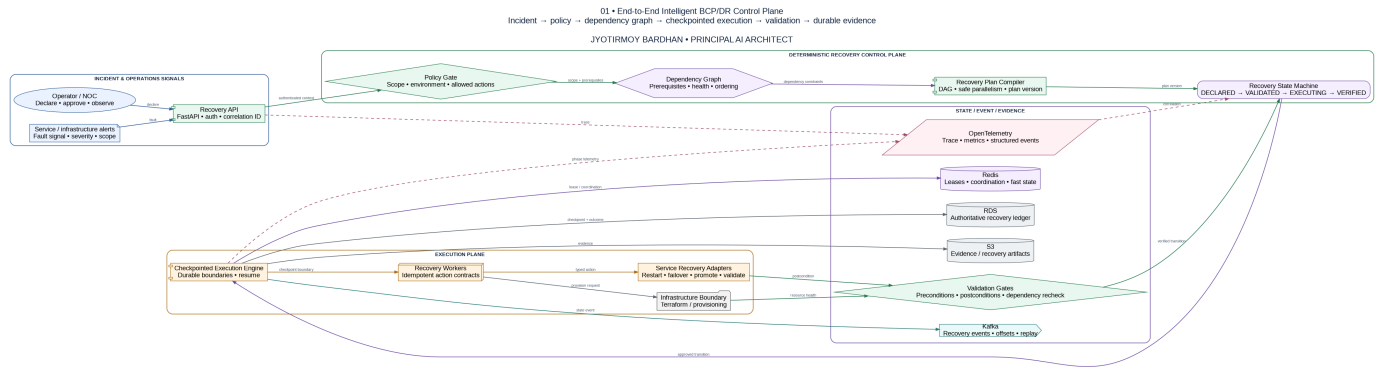
02 • ARCHITECTURE EVIDENCE MAP

The architecture is traceable from mechanism to control boundary to production evidence.

Mechanism	Control	Evidence
Dependency-aware state	Explicit prerequisite ordering	Dependency-chain validation
Checkpointing	Durable safe resume point	Checkpointed execution
Idempotency	Replay-safe action contract	Idempotent recovery actions
Failure isolation	Bounded blast radius	Failure-domain isolation
Deterministic authority	Explicit state advancement	Separated recovery authority
Validation	Advance only on proof	95%+ automated validation
Resilience outcome	Continuity objective	Sub-5-min RTO / near-zero RPO / zero data loss / 99%+ success

01 • End-to-End Intelligent BCP/DR Control Plane

Incident → policy → dependency graph → checkpointed execution → validation → durable evidence



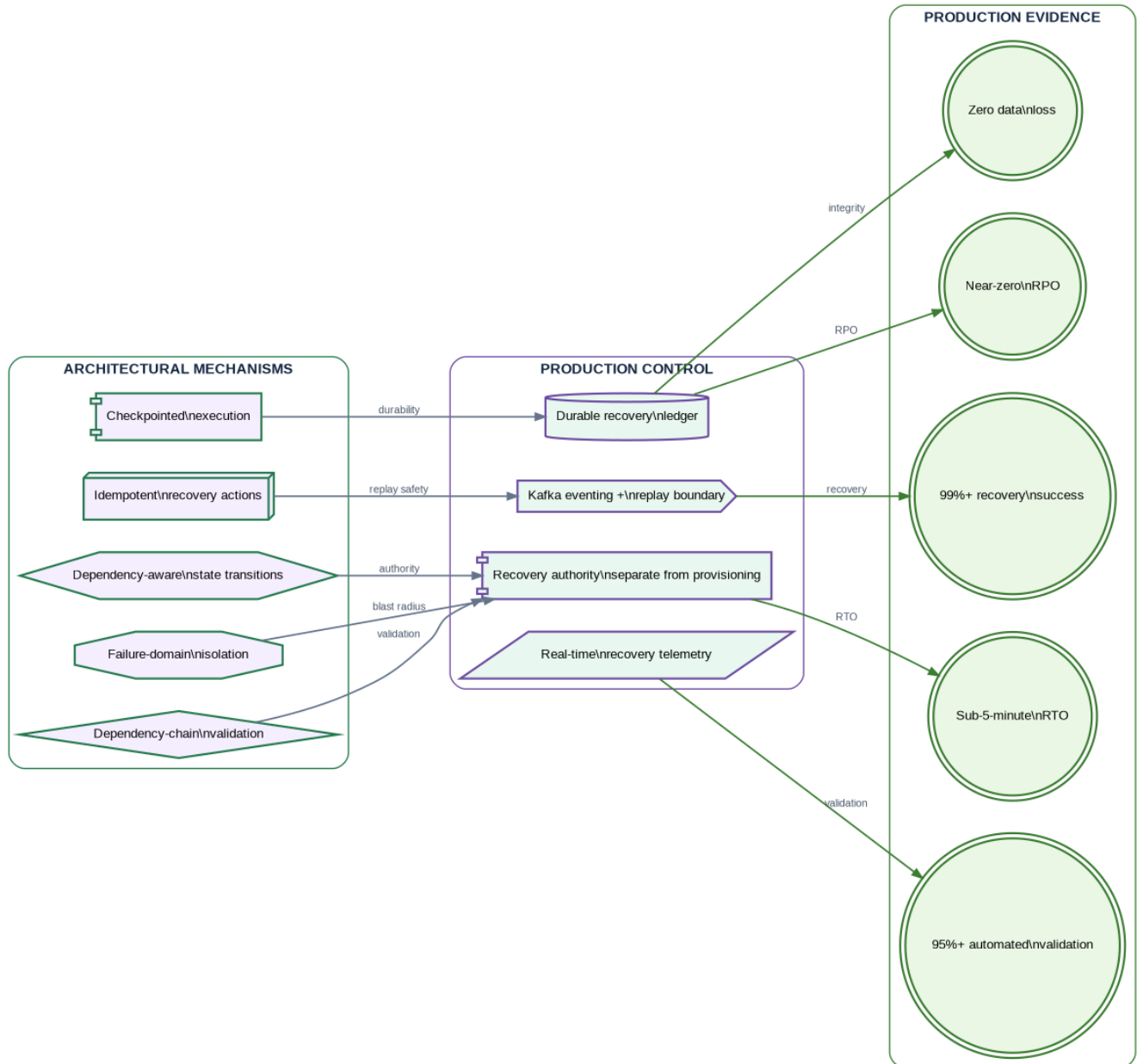
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02 • Architecture Evidence Map

Mechanism → control boundary → measurable production evidence

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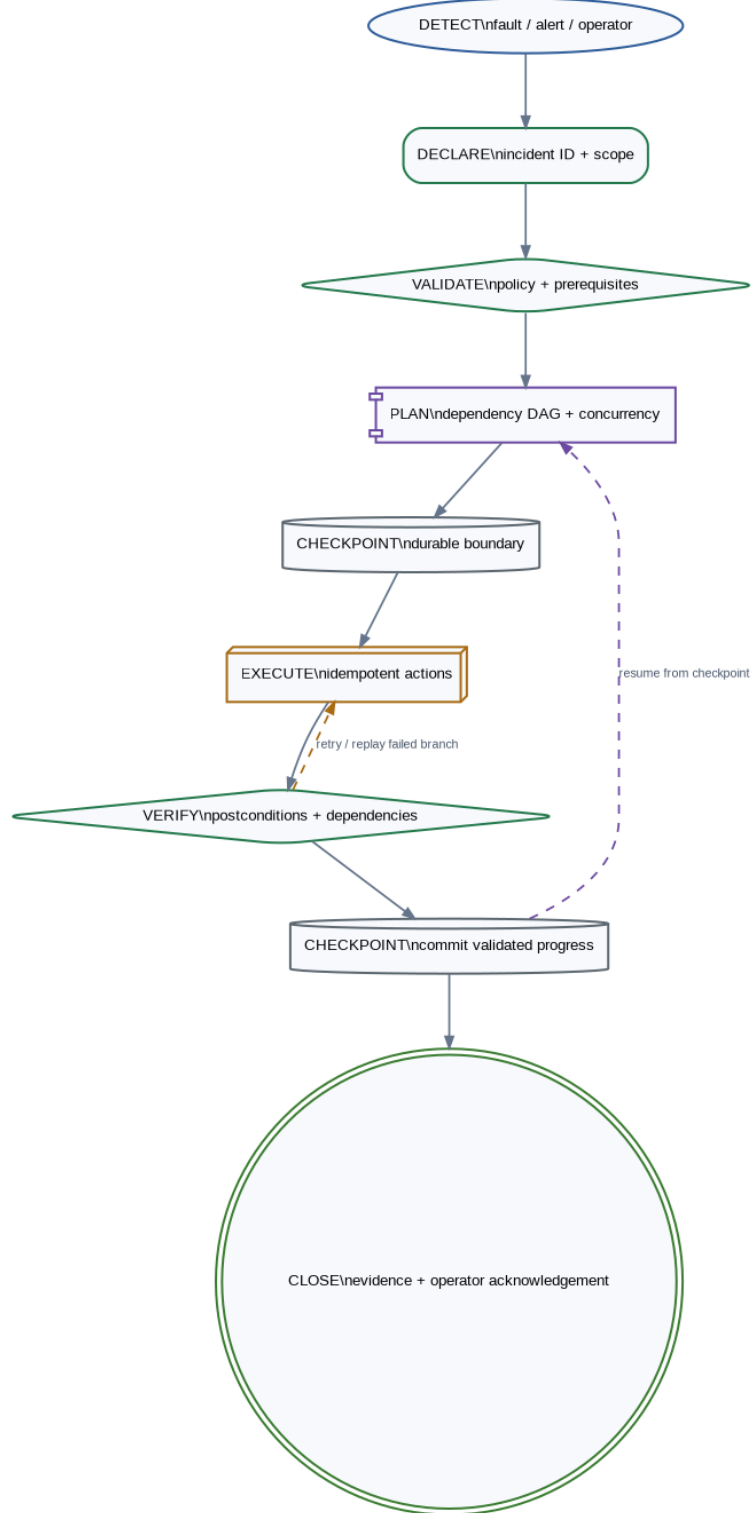
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03 • Business Continuity Control Loop

A recovery incident is a state machine, not a flat runbook

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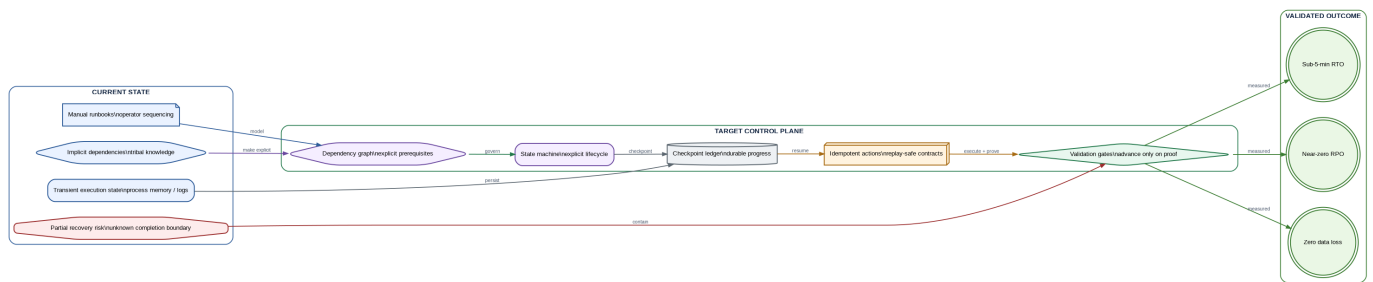


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04 • Current-State → Target-State Transformation

From implicit runbooks and fragmented state to an explicit recovery control plane

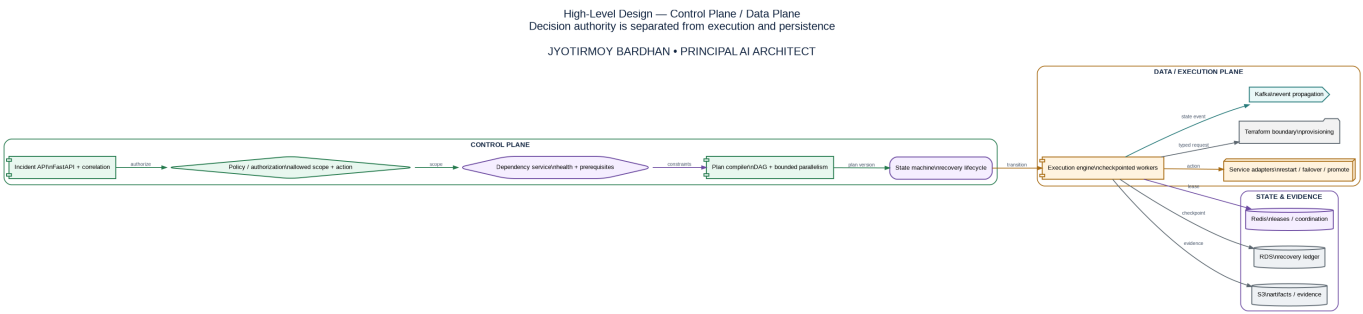
Current-State → Target-State Transformation
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05 • High-Level Design — Control Plane / Data Plane

Decision authority is separated from execution and persistence



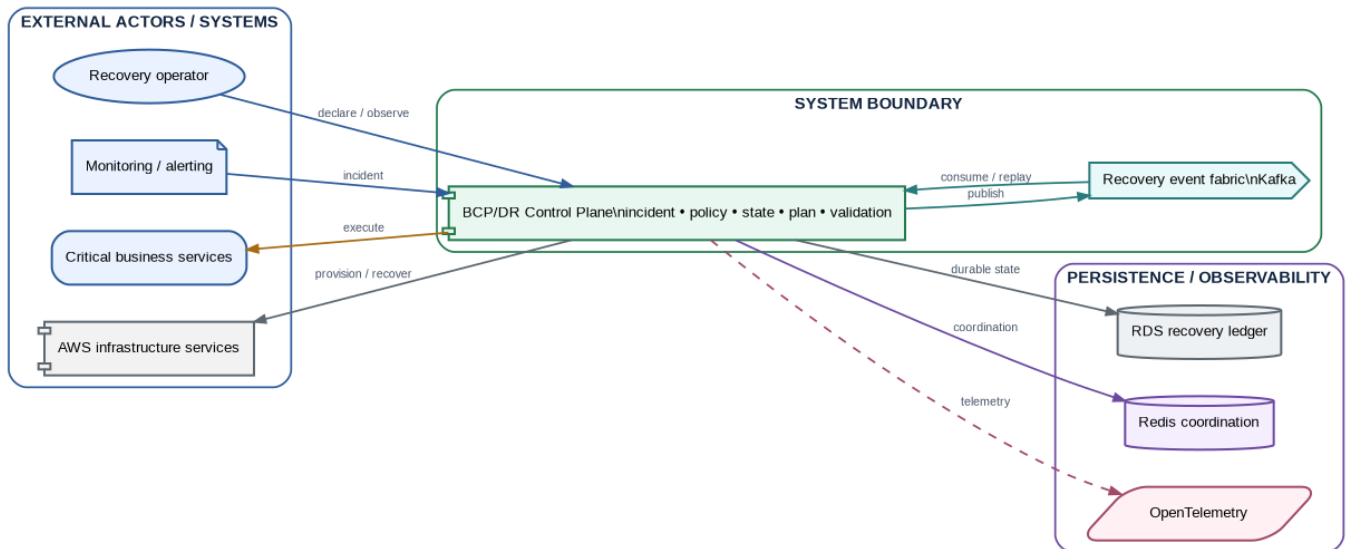
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06 • C4 Context — Recovery Control Plane

External actors and systems around the architectural boundary

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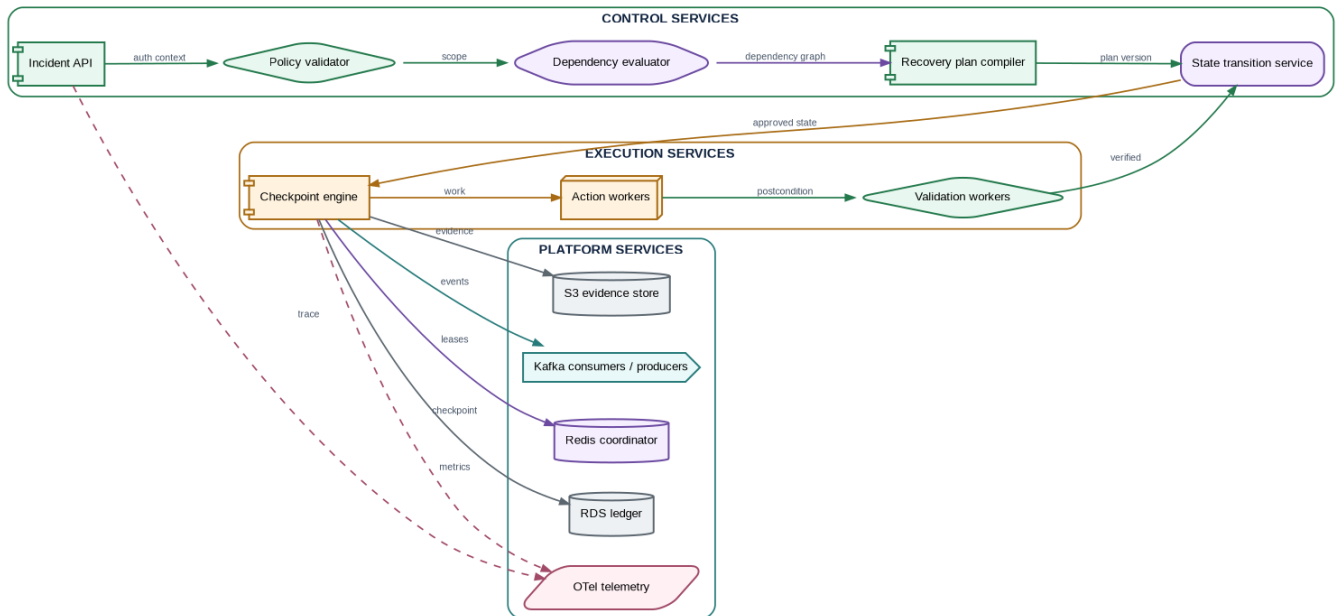
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07 • C4 Container — Internal Recovery Services

Container-level decomposition of the control plane

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Container-level decomposition of the control plane

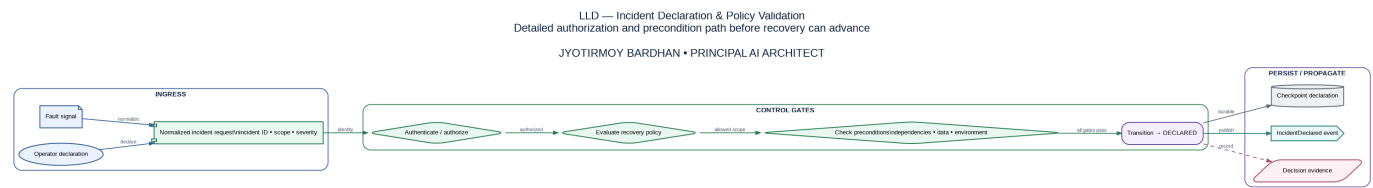
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08 • LLD — Incident Declaration & Policy Validation

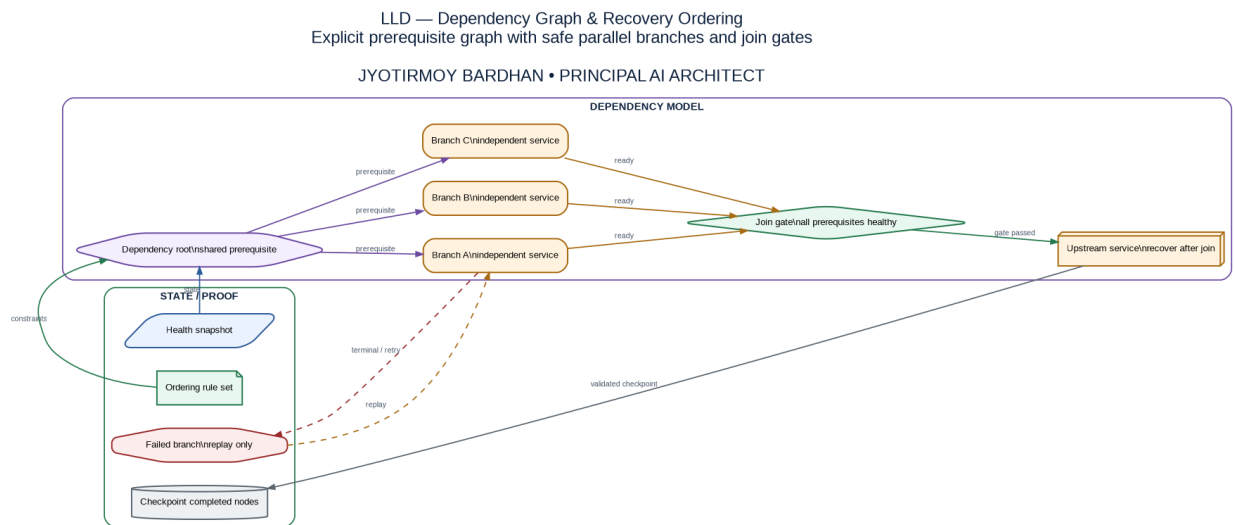
Detailed authorization and precondition path before recovery can advance



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09 • LLD — Dependency Graph & Recovery Ordering

Explicit prerequisite graph with safe parallel branches and join gates



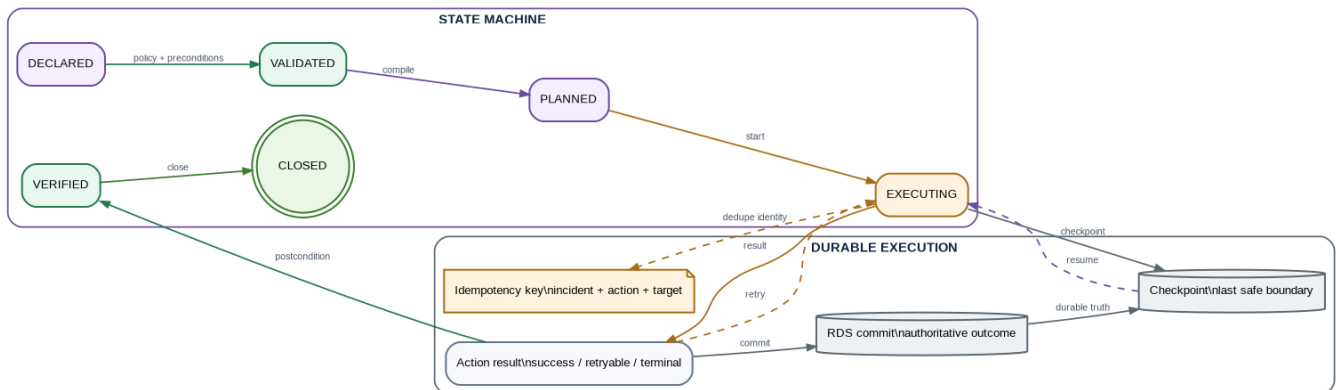
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10 • LLD — Checkpointed Execution Engine

Durable state transitions, replay boundaries and worker interruption handling

LLD — Checkpointed Execution Engine
Durable state transitions, replay boundaries and worker interruption handling

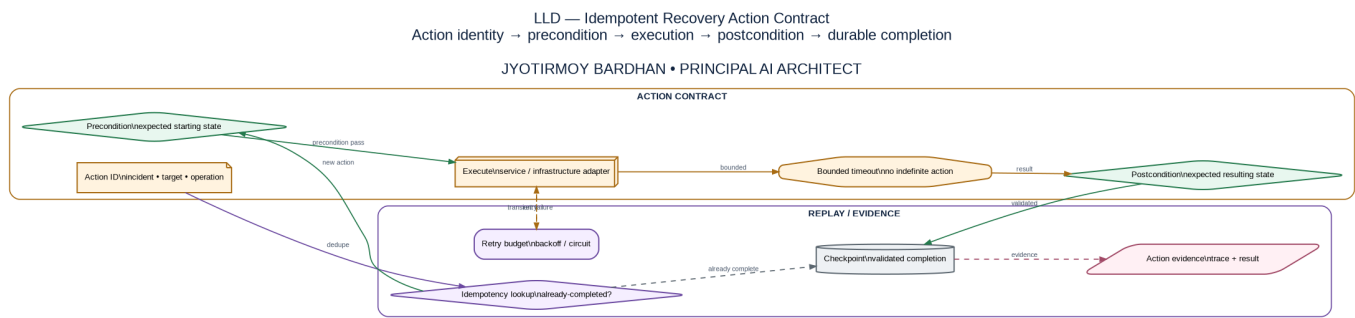
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11 • LLD — Idempotent Recovery Action Contract

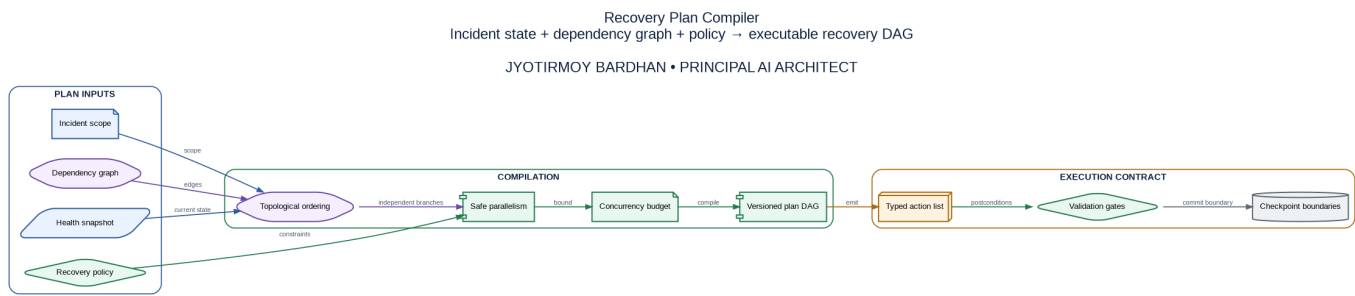
Action identity → precondition → execution → postcondition → durable completion



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12 • Recovery Plan Compiler

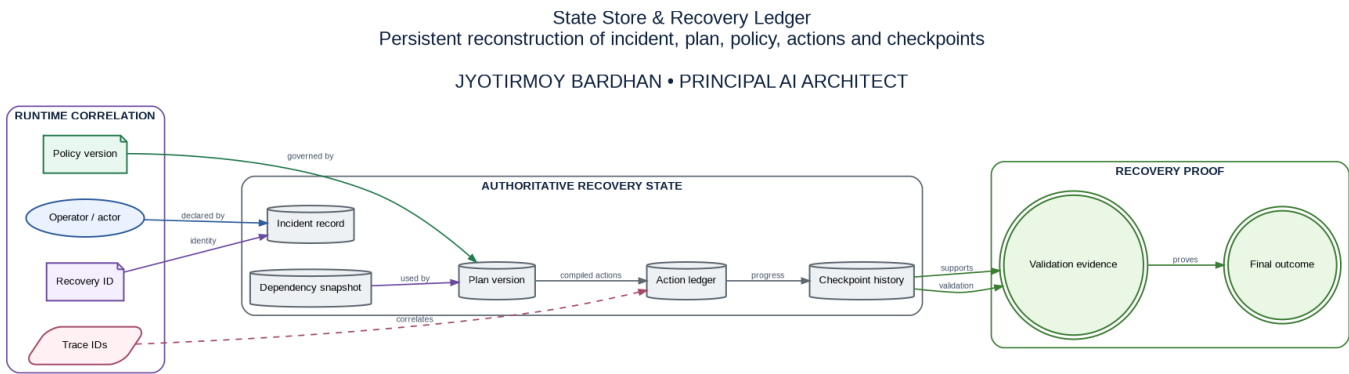
Incident state + dependency graph + policy → executable recovery DAG



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13 • State Store & Recovery Ledger

Persistent reconstruction of incident, plan, policy, actions and checkpoints



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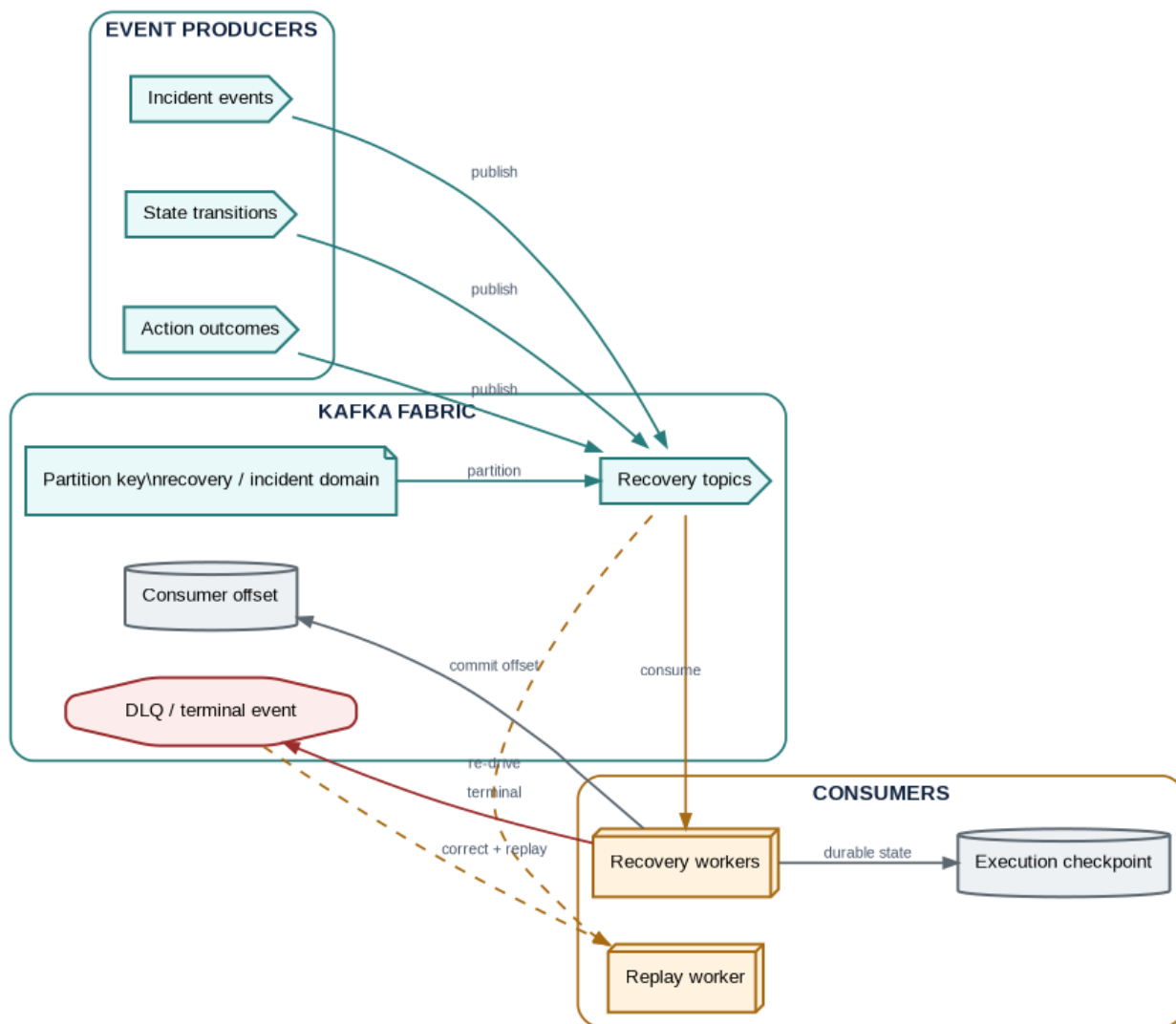
14 • Event-Driven Recovery Fabric

Kafka propagation, consumer isolation, offsets, replay and durable state alignment

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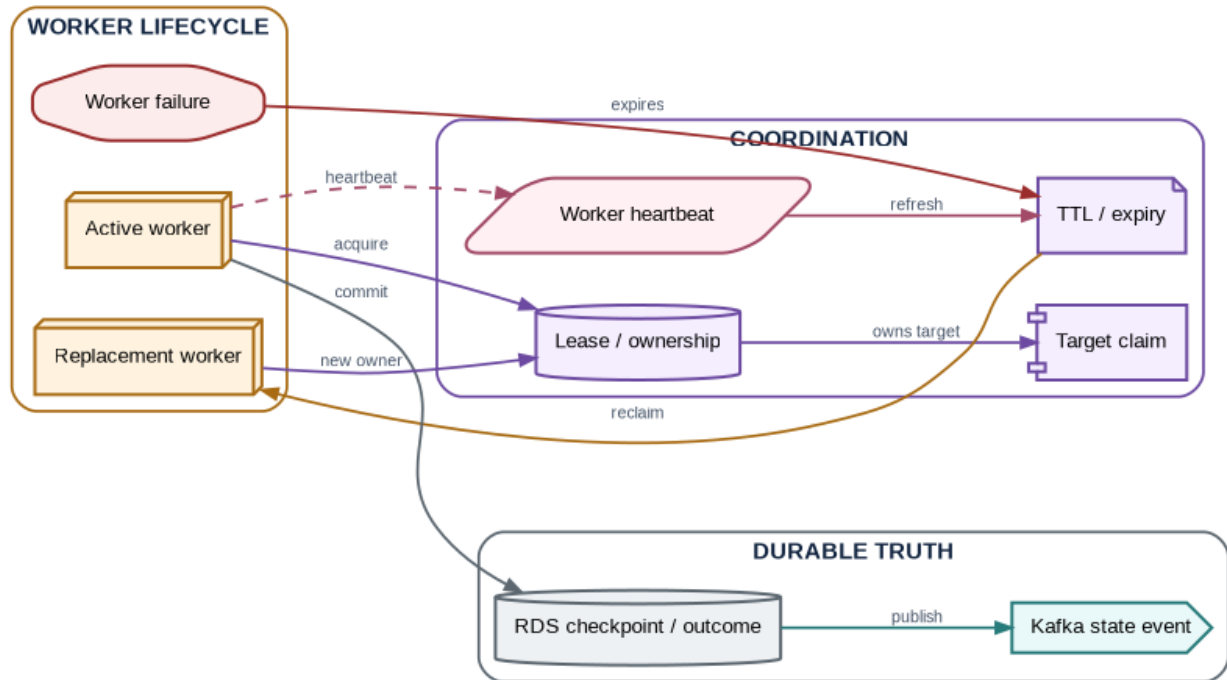
15 • Redis Coordination & Fast Recovery State

Leases, ownership, heartbeat, TTL and handoff around durable recovery state

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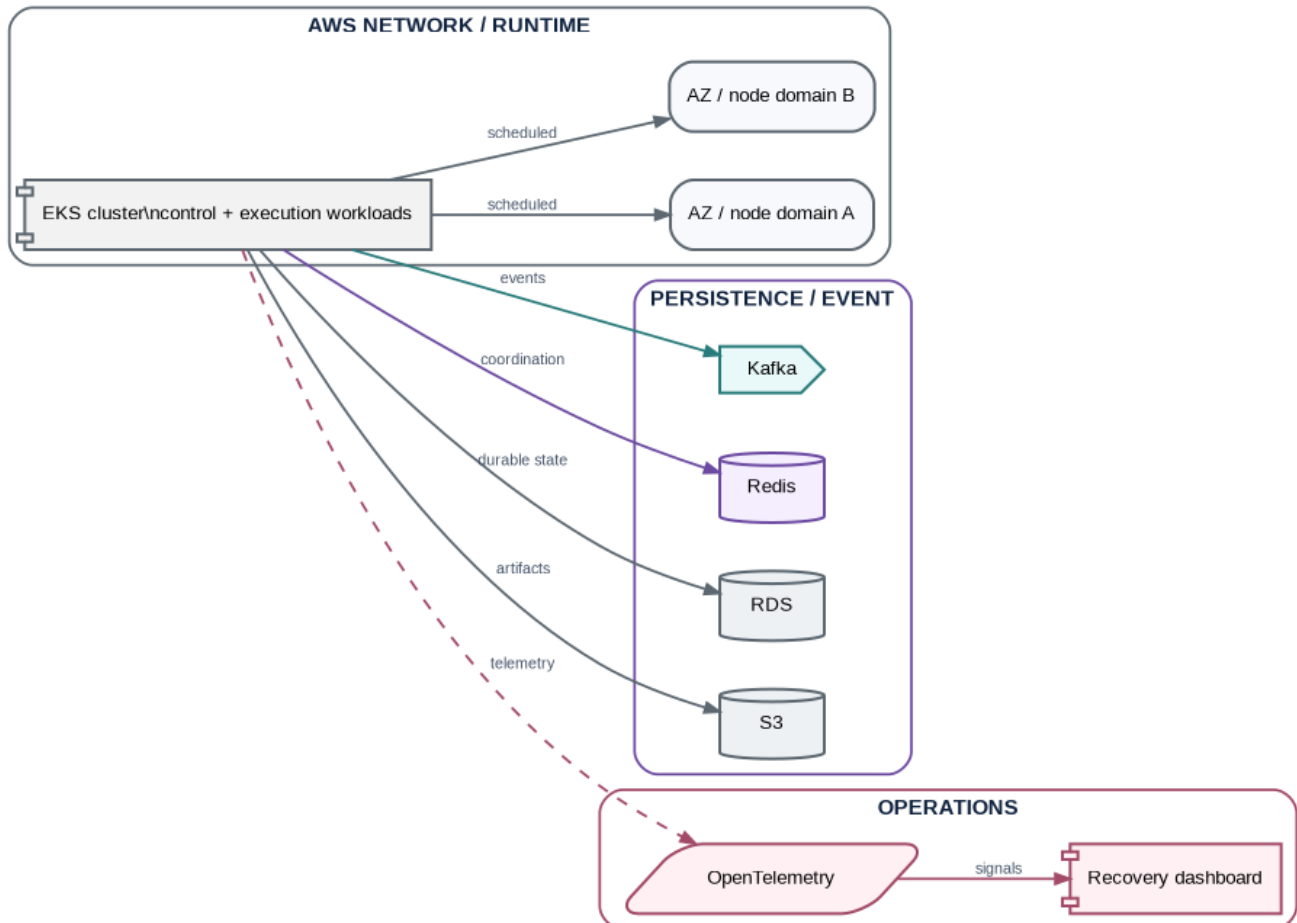
16 • AWS Deployment Topology

EKS runtime with durable state, eventing, coordination and observability boundaries

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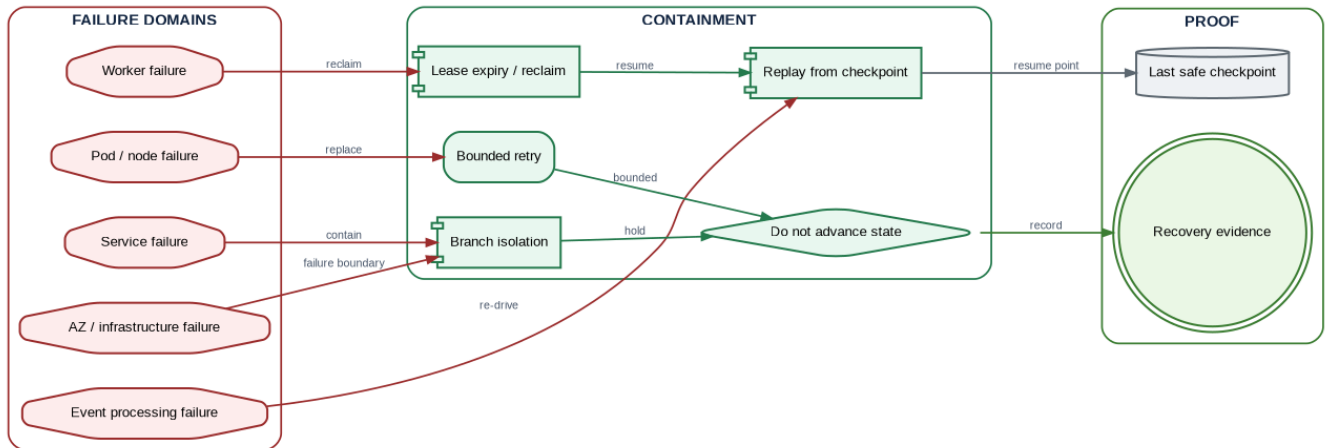
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17 • Failure Domains & Blast-Radius Isolation

Worker, pod/node, service, event and infrastructure failures are isolated by boundary

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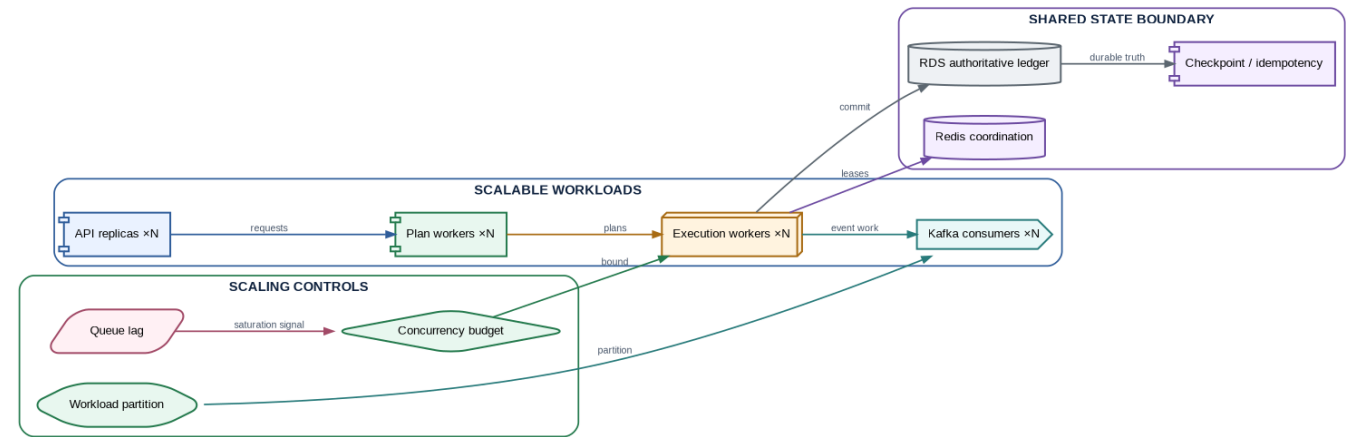


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18 • Horizontal Scaling & Workload Partitioning

Scale stateless control and execution workers while keeping recovery state authoritative

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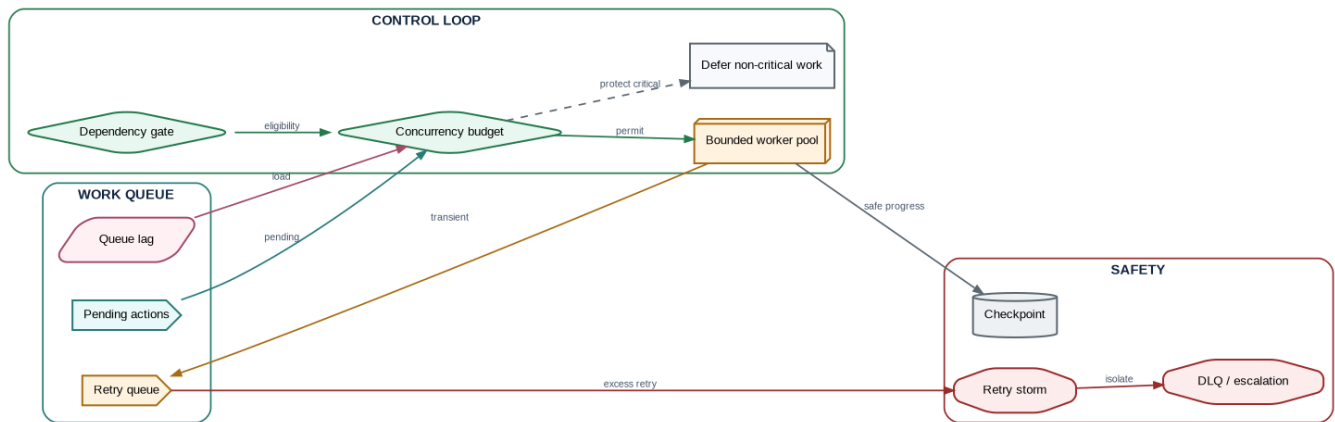
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19 • Recovery Backpressure & Concurrency Control

Prevent recovery storms and unsafe parallel execution

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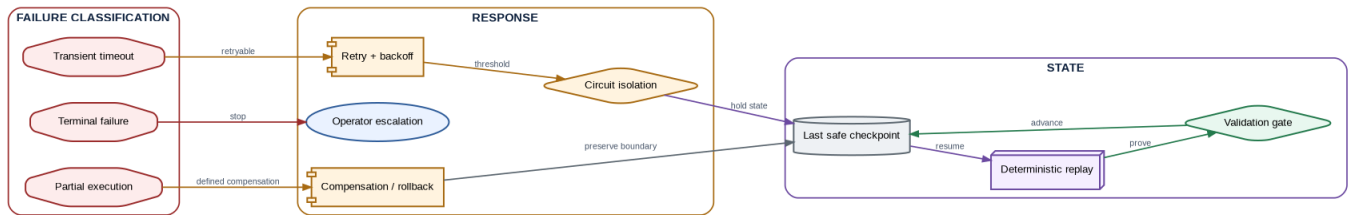
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20 • Resilience Patterns — Retry / Circuit / Compensation

Failure-class-aware recovery without corrupting recovery state

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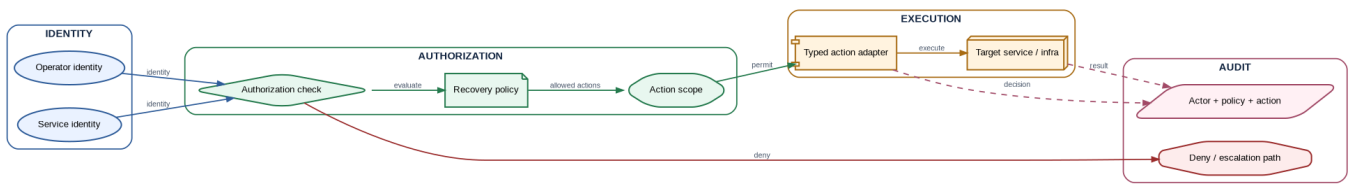


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21 • Zero-Trust Recovery Execution Boundary

Identity → policy → least privilege → typed action → audit

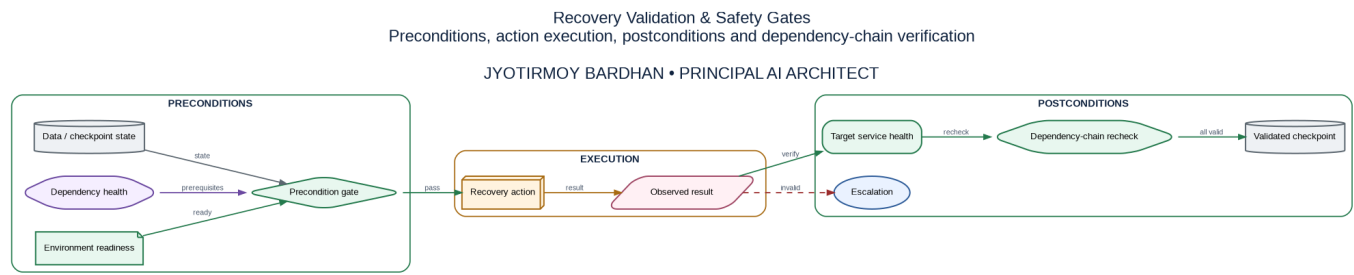
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22 • Recovery Validation & Safety Gates

Preconditions, action execution, postconditions and dependency-chain verification



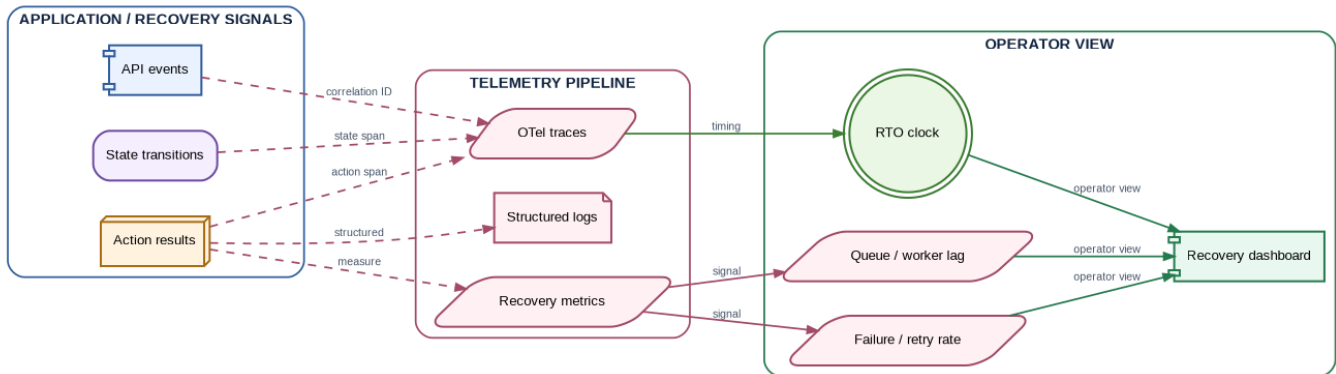
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23 • Observability Architecture

Incident-wide correlation across declaration, planning, execution and validation

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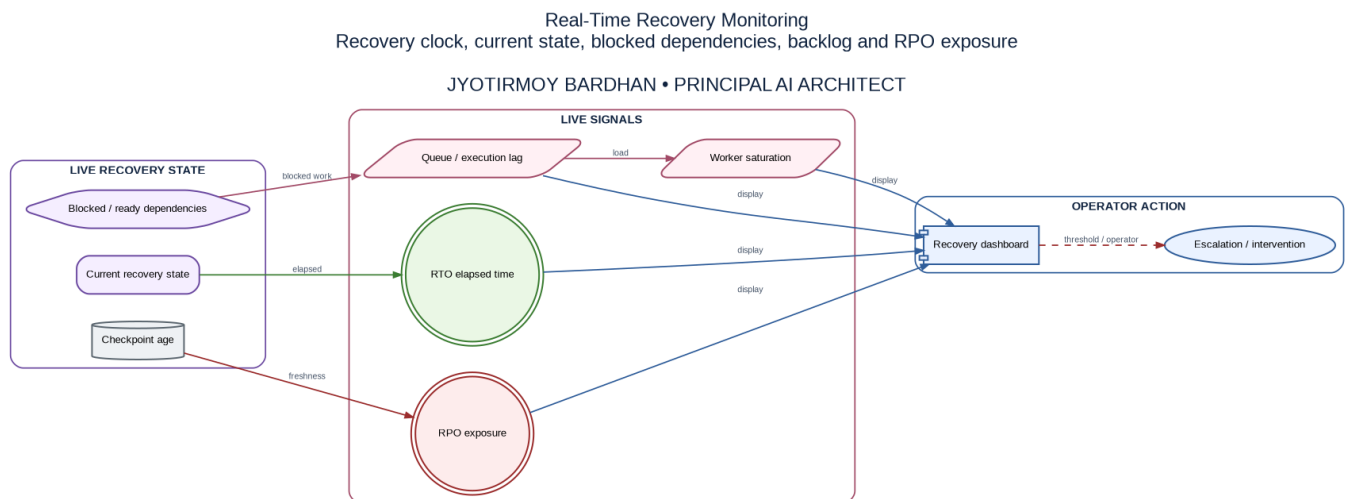
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24 • Real-Time Recovery Monitoring

Recovery clock, current state, blocked dependencies, backlog and RPO exposure

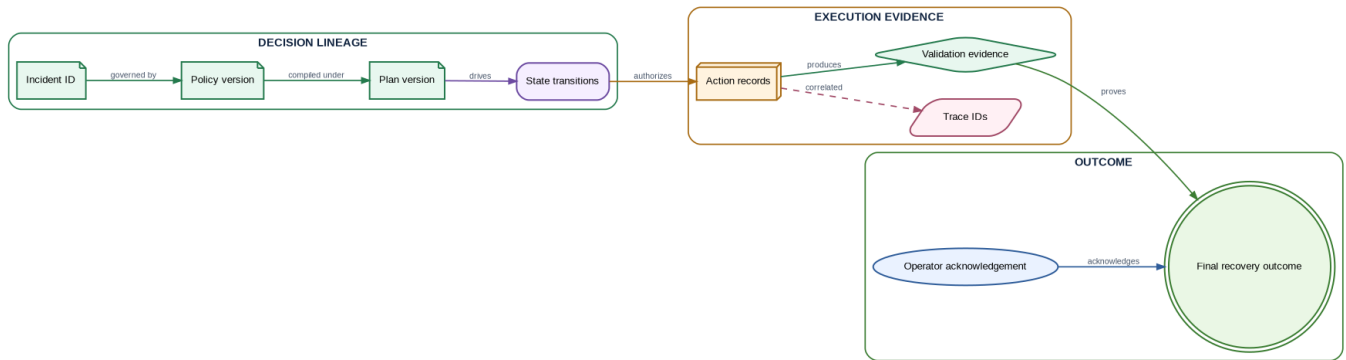


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25 • Recovery Auditability & Decision Lineage

Incident → policy → plan → action → validation → outcome

Recovery Auditability & Decision Lineage
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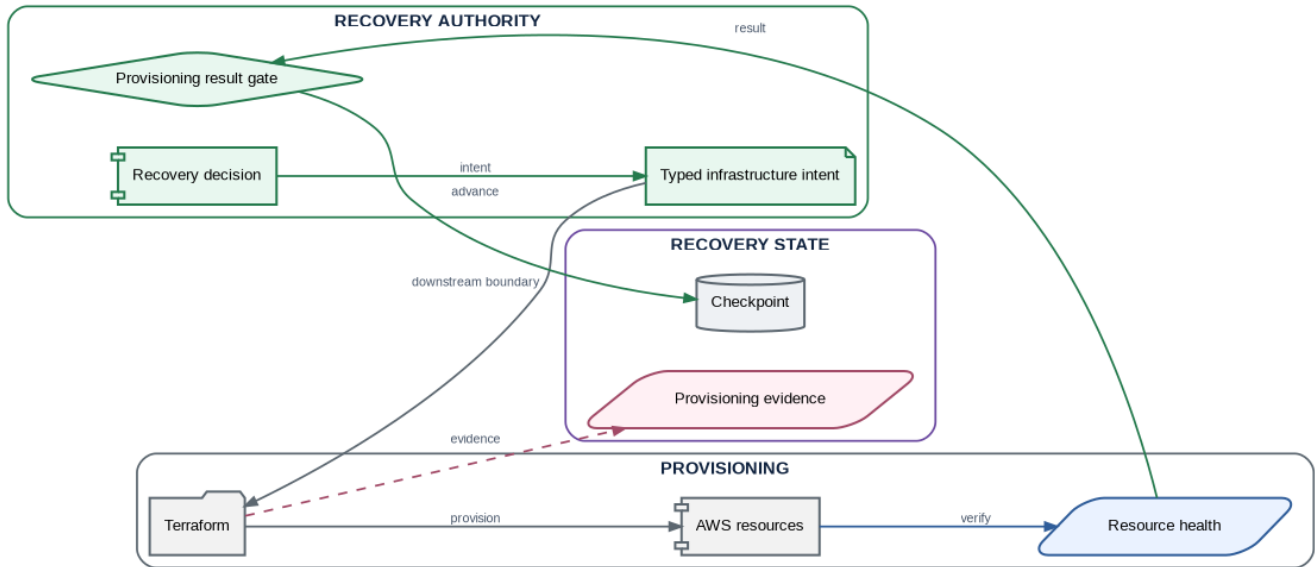
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26 • Infrastructure Provisioning Boundary

Recovery authority requests typed infrastructure actions; Terraform remains the provisioning boundary

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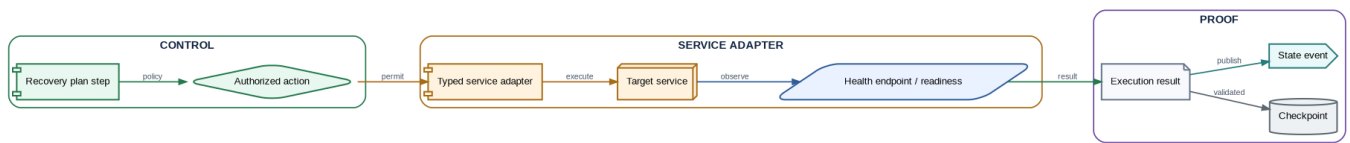
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27 • Service Execution Boundary

Control-plane decisions are translated into typed downstream service actions

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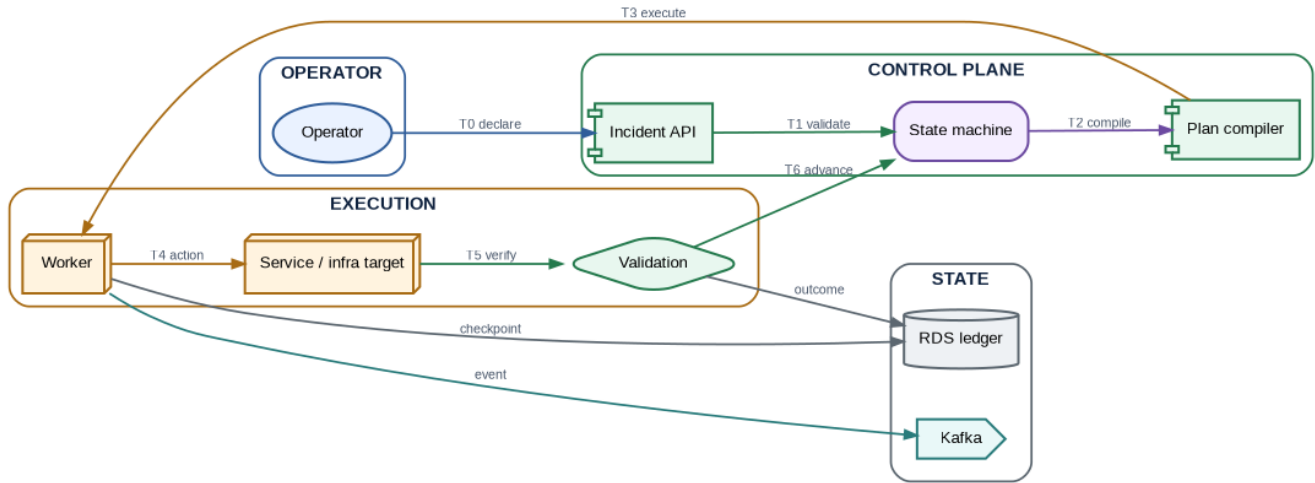
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28 • Disaster Recovery Sequence — Full Incident

Sequence of control decisions, execution and proof across the recovery lifecycle

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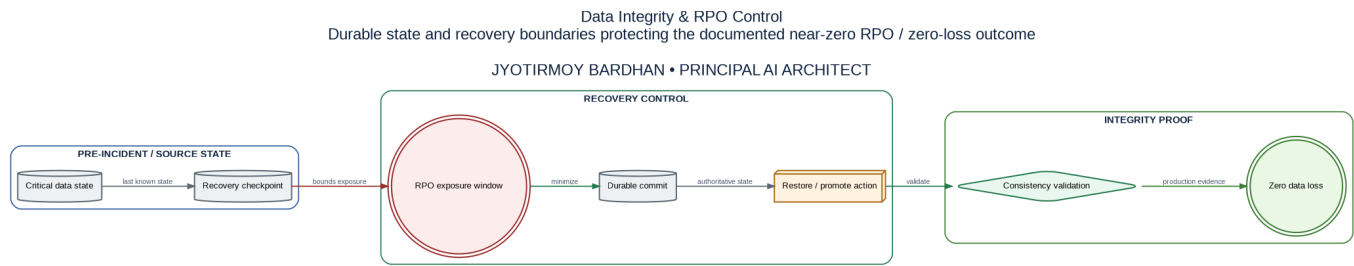
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29 • Data Integrity & RPO Control

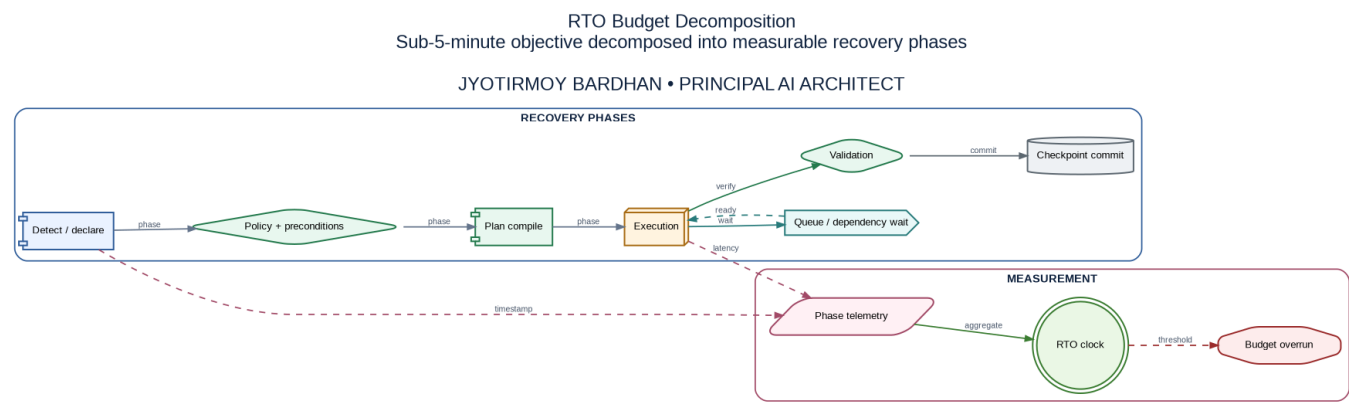
Durable state and recovery boundaries protecting the documented near-zero RPO / zero-loss outcome



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30 • RTO Budget Decomposition

Sub-5-minute objective decomposed into measurable recovery phases



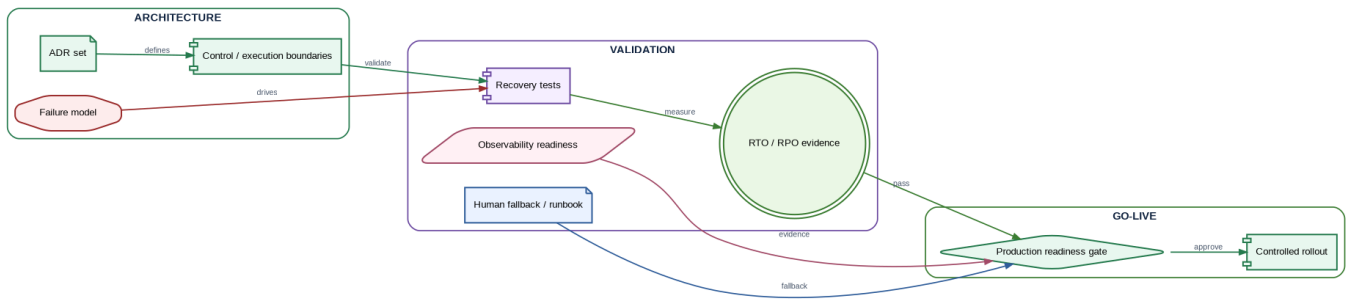
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31 • Production Readiness Gate

Evidence required before recovery automation is trusted operationally

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Evidence required before recovery automation is trusted operationally

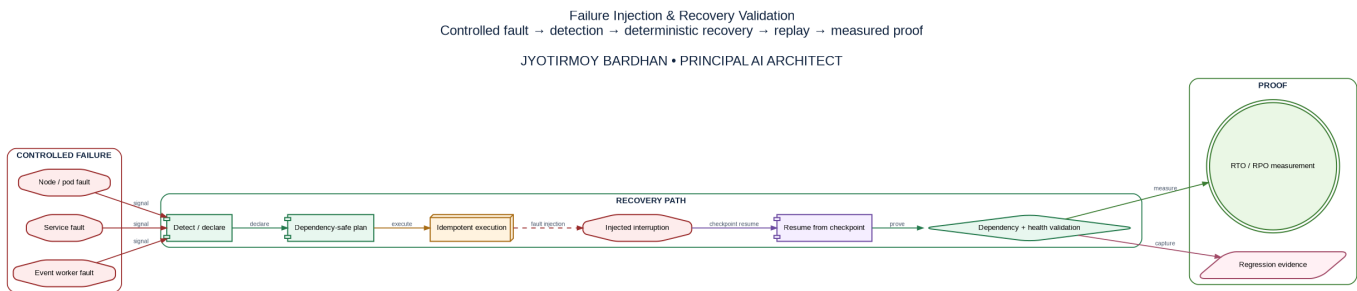
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32 • Failure Injection & Recovery Validation

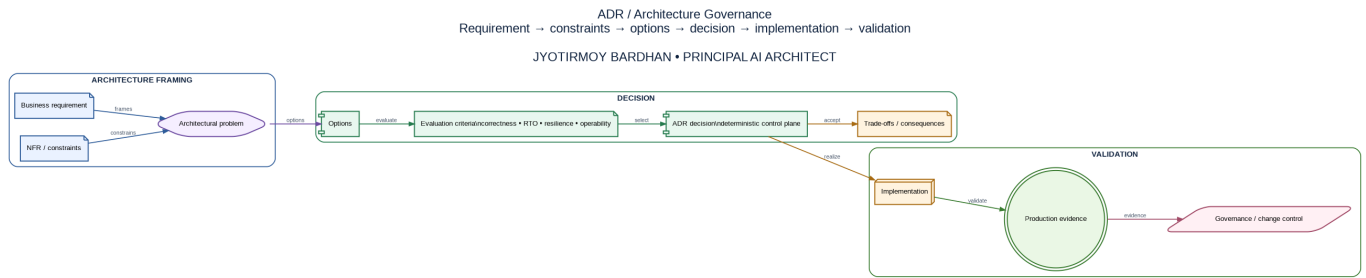
Controlled fault → detection → deterministic recovery → replay → measured proof



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33 • ADR / Architecture Governance

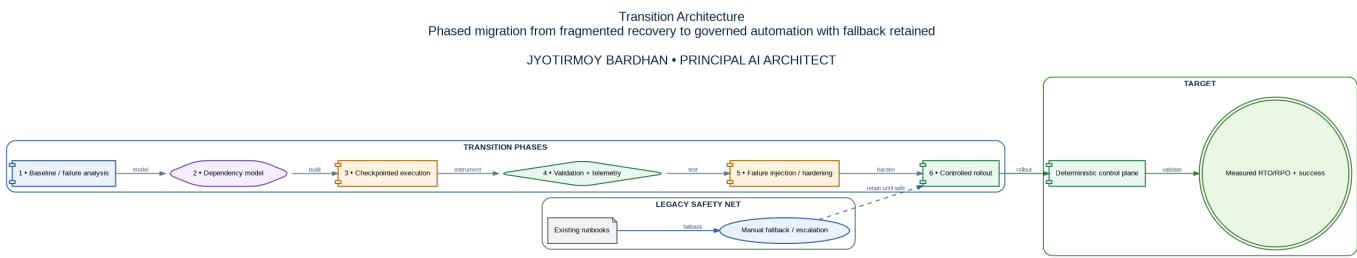
Requirement → constraints → options → decision → implementation → validation



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34 • Transition Architecture

Phased migration from fragmented recovery to governed automation with fallback retained



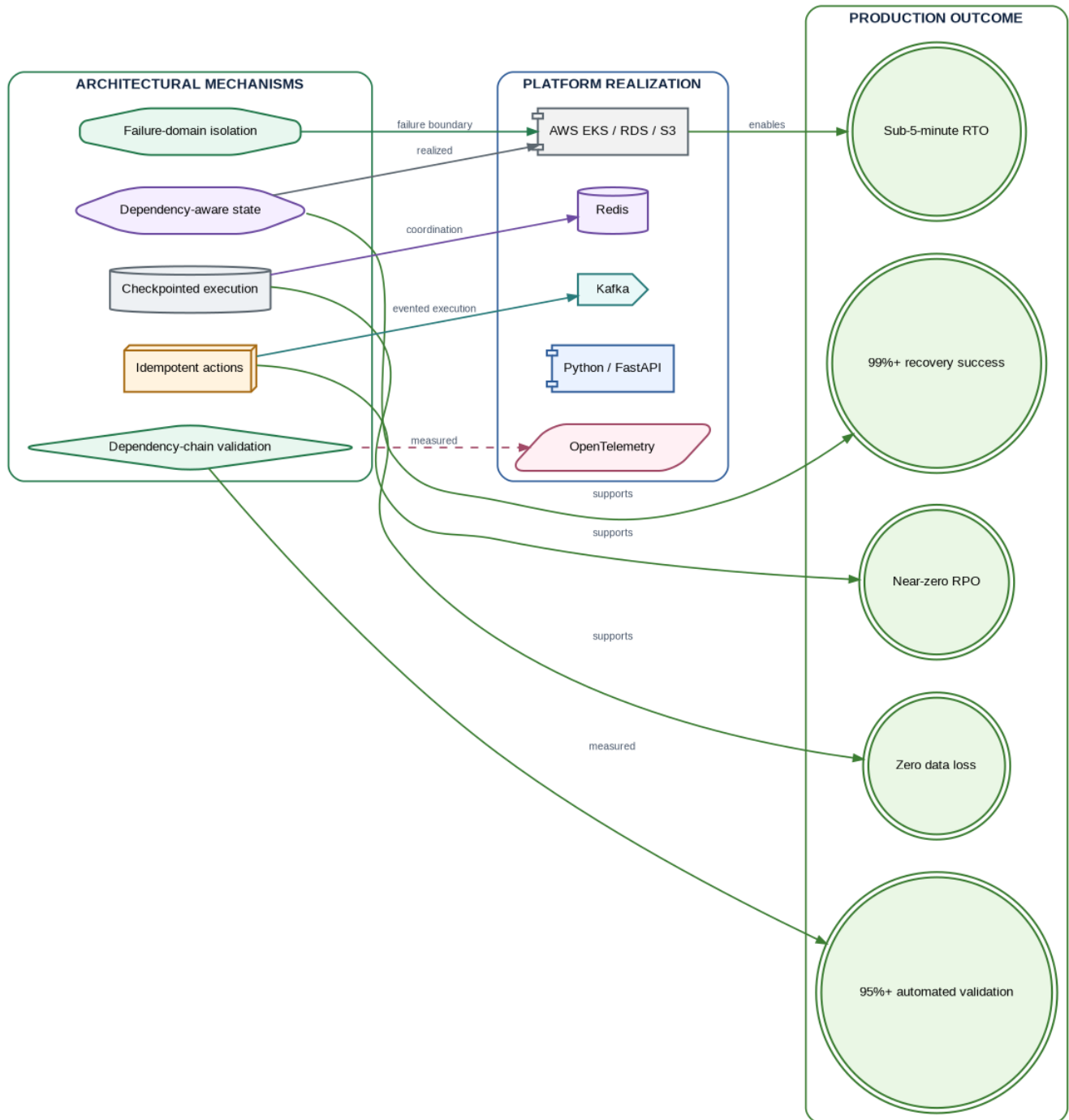
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35 • Production Validation & Outcome Chain

Architecture mechanism → production behavior → resilience outcome

Production Validation & Outcome Chain
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39 • Problem Framing

BCP/DR recovery is a distributed correctness problem. A fast sequence can still produce inconsistent state when dependencies are implicit, actions are replayed unsafely or completion is inferred from transient process state.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

40 • Real-World Constraints

The architecture had to preserve downstream service and infrastructure execution boundaries while improving recovery determinism, dependency ordering, RTO/RPO behavior, partial-failure handling, replay safety and auditability.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

41 • Non-Functional Requirements

The documented NFR posture centers on sub-5-minute RTO, near-zero RPO, zero-data-loss behavior, deterministic state progression, bounded blast radius, replay safety and measurable recovery evidence.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

42 • Current-State Failure Analysis

Fragmented recovery creates hidden state: completed work, unmet prerequisites and safe resume points are difficult to represent consistently. The target makes those facts explicit.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

43 • Target-State Architecture

The target introduces recovery authority, dependency state, plan compilation, checkpointed execution, idempotent action contracts, validation gates and durable evidence.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

44 • Dependency-Aware Recovery

Recovery is represented as a dependency graph. Independent branches can execute within bounded concurrency; upstream transitions wait for prerequisite validation.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

45 • Checkpointed Execution

A checkpoint is a durable recovery boundary. An action attempt is not equivalent to a committed recovery state; validation precedes advancement.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

46 • Idempotency Model

Action identity links incident, target and operation so retries or worker replacement can safely replay without duplicating completed semantic effects.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

47 • Event-Driven Recovery

Kafka decouples event propagation and supports replay. Authoritative recovery state remains durable and explicit.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

48 • Recovery Ledger

Durable state captures incident lifecycle, plan identity, dependency snapshot, action outcomes, checkpoints and validation evidence.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

49 • AWS Deployment

The documented runtime uses AWS EKS for containerized workloads, Kafka for events, Redis for coordination, RDS for durable recovery state and S3 for artifacts/evidence.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

50 • Failure-Domain Isolation

Worker, pod/node, service, event-stream and infrastructure failures are treated as separate domains so local faults do not automatically advance unrelated recovery branches.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

51 • Scalability Strategy

Stateless control and execution workloads can scale horizontally. Stateful coordination and durable stores remain explicit boundaries; worker count is constrained by safe recovery concurrency.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

52 • Backpressure & Recovery Storm Control

Queue lag, dependency readiness, worker capacity and retry budgets constrain recovery storms and protect critical branches.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

53 • Resilience Engineering

Retry, backoff, circuit isolation, checkpointing, replay and escalation are selected by failure class.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

54 • Security & Recovery Authorization

Recovery operations are privileged. Identity, authorization, policy and execution are separated so an incident does not imply unrestricted action authority.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

55 • Recovery Validation

Preconditions determine whether an action may start; postconditions determine whether it succeeded; dependency rechecks determine whether the broader graph remains consistent.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

56 • Observability Architecture

OpenTelemetry correlates incident declaration, planning, execution and validation so recovery latency and distributed failures can be diagnosed.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

57 • Real-Time Recovery Monitoring

The operator view centers on recovery state, RTO clock, blocked dependencies, action backlog, checkpoint age, worker saturation and RPO exposure.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

58 • Auditability & Decision Lineage

Incident ID, policy version, plan version, state transitions, action identity, validation evidence and trace identifiers form a reconstructable chain.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

59 • Infrastructure Provisioning Boundary

Terraform remains a downstream provisioning boundary. Recovery authority emits typed infrastructure intent instead of embedding provisioning logic into recovery state management.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

60 • Service Execution Boundary

Service adapters translate deterministic recovery decisions into downstream operations and return evidence. The control plane decides whether that evidence is sufficient to advance.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

61 • RTO Budget Decomposition

The sub-5-minute objective is a budget across detection, validation, planning, execution, dependency waiting, verification and checkpoint commit.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

62 • RPO & Data Integrity

Near-zero RPO requires durable state boundaries. Recovery progress cannot depend solely on transient worker memory.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

63 • Failure Injection & Validation

Controlled failures validate dependency handling, worker interruption, replay, checkpoint resume, idempotency and post-recovery consistency.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

64 • ADR Governance

Architecture decisions follow requirement → constraints → problem → options → criteria → decision → trade-offs → implementation → validation.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

65 • Transition Architecture

Migration proceeds through baseline analysis, dependency modeling, checkpointed execution, validation/telemetry, hardening and controlled rollout while fallback procedures remain available.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.

66 • Principal Contribution & Production Outcome

The documented architectural contribution is dependency-aware state transitions, checkpointed execution, idempotent recovery actions, failure-domain isolation and separation from provisioning/service execution. Documented outcomes: sub-5-minute RTO, near-zero RPO, zero data loss, 95%+ automated recovery validation and 99%+ recovery success.

Decision lens

Correctness under partial failure • replay safety • bounded blast radius • deterministic state progression • measurable recovery evidence.